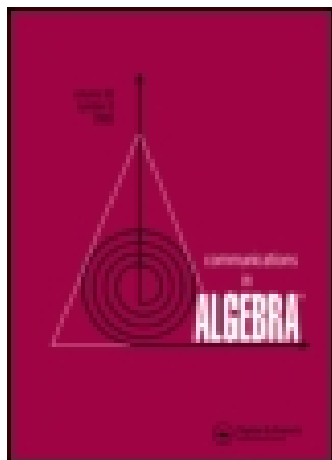




## Communications in Algebra

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/lagb20>

### Geometric hyperplanes in generalized hexagons of order (2,2)

Daniel Frohardt<sup>a</sup> & Peter Johnson<sup>b</sup>

<sup>a</sup> Department of Mathematics , Wayne State University , Detroit, MI, 48202, USA

<sup>b</sup> Department of Mathematics , Kansas State University , Manhattan, KS, 66506, USA

Published online: 27 Jun 2007.

To cite this article: Daniel Frohardt & Peter Johnson (1994) Geometric hyperplanes in generalized hexagons of order (2,2), Communications in Algebra, 22:3, 773-797, DOI: [10.1080/00927879408824875](https://doi.org/10.1080/00927879408824875)

To link to this article: <http://dx.doi.org/10.1080/00927879408824875>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at <http://www.tandfonline.com/page/terms-and-conditions>

# GEOMETRIC HYPERPLANES IN GENERALIZED HEXAGONS OF ORDER (2,2)

Daniel Frohardt

Department of Mathematics  
Wayne State University  
Detroit, MI 48202, USA

Peter Johnson

Department of Mathematics  
Kansas State University  
Manhattan, KS 66506, USA

**Abstract:** We present a detailed study of the geometric hyperplanes of the two generalized hexagons of order  $(2, 2)$ . This leads to concrete descriptions of the universal embeddings of these hexagons, as well as a description of the  $G_2(2)$ -orbits on the Lie algebra  $\mathfrak{g}_2(2)$ , illustrating some of the anomalies of this algebra. As a byproduct of our investigations, we develop some general theory that can be applied to other incidence systems with 3 points per line.

## 1. Introduction

The utility and elegance of Tits buildings for the study of Chevalley groups is well established. In this paper we use geometrical properties of the two buildings for  $G = G_2(2)$  to represent two 14-dimensional  $\mathbb{F}_2G$ -modules (the adjoint module and a certain reducible uniserial module) in a concrete fashion. This makes it easy to see how the  $G$ -orbits intersect given subspaces of the modules and to obtain other related information. As an illustration, we show how the geometry makes plain some anomalous features of the associated Lie algebra  $\mathfrak{g}_2(2)$ . Our methods can be used to study many other group modules over  $\mathbb{F}_2$ .

In general, we use standard notation for groups and geometries, as found in the Atlas [3] and Dembowski [4]. Let  $S = (\mathcal{P}, \mathcal{L})$  be an incidence system of points and lines, where lines are regarded as sets of points. A subset  $H$  of  $\mathcal{P}$  is a *geometric hyperplane* of  $S$  if every line not contained in  $H$  meets  $H$  in a single point. Let  $V(S)$  be the collection of all geometric hyperplanes of  $S$ , including the "degenerate" one,  $\mathcal{P}$  itself. We assume throughout that

every line of  $\mathcal{S}$  has exactly three points. Then, as explained later,  $V(\mathcal{S})$  has the natural structure of an  $\mathbf{F}_2\text{Aut}(\mathcal{S})$ -module.

Our primary goal is to provide a detailed study of the properties of  $V(\mathcal{S})$  when  $\mathcal{S}$  is a hexagon of order  $(2, 2)$ . (Definitions appear below.) These hexagons are buildings of type  $G_2$ , with automorphism group  $G_2(2)$ , by results of Tits (see Cohen and Tits [2]). Throughout this paper,  $\mathcal{H}$  will denote the “classical”  $G_2(2)$  hexagon, and  $\mathcal{H}^*$  will denote its dual. Explicit descriptions of these hexagons can be found in [2].

Following Ronan [9], an *embedding* of  $\mathcal{S}$  is a map  $\phi : \mathcal{P} \mapsto W$ , where  $W$  is a vector space, such that  $\phi$  maps every line of  $\mathcal{S}$  bijectively onto a projective line of  $W$ . An embedding  $\phi : \mathcal{P} \mapsto U$  is a *universal embedding* if every embedding  $\psi : \mathcal{P} \mapsto W$  can be factored as  $f \circ \phi$  for some semilinear map  $f : U \mapsto W$ . Our assumption that  $\mathcal{S}$  has three points per line implies that whenever  $\mathcal{S}$  has an embedding, it has a universal embedding (unique up to isomorphism) into a vector space  $U = U(\mathcal{S})$ . Here  $U(\mathcal{S})$  is a vector space over  $\mathbf{F}_2$  generated by vectors  $v_p$ ,  $p \in \mathcal{P}$ , subject only to the relations  $v_p + v_q + v_r = 0$  whenever  $\{p, q, r\} \in \mathcal{L}$ . It is an  $\mathbf{F}_2\text{Aut}(\mathcal{S})$ -module.

Some general theory is developed in Section 2. In particular, we describe the natural isomorphism between the dual  $U(\mathcal{S})^*$  of the universal embedding space and the module  $V(\mathcal{S})$  of geometric hyperplanes. This result, which is fairly well-known, allows us to shift the focus from universal embeddings to the concrete situation of modules whose vectors are geometric hyperplanes. We also examine structures (such as group-invariant quadratic forms and algebra operations) that can be defined geometrically on  $U(\mathcal{S})$  or  $V(\mathcal{S})$ .

Section 3 summarizes known properties of the hexagons of order  $(2, 2)$ . These are used to obtain a new proof that the universal embedding spaces are 14-dimensional and, in the case of  $\mathcal{H}^*$ , to obtain technical results which are needed elsewhere. In Sections 4 and 5 we present detailed descriptions of the modules consisting of the geometric hyperplanes of  $\mathcal{H}$  and  $\mathcal{H}^*$ , respectively. The latter module supports a Lie algebra  $\mathfrak{g}_2(2)$  whose connections with the geometry of  $\mathcal{H}^*$  are the subject of Section 6.

Many of our results were stated without proof in [6], and [7] contains related investigations. The authors would like to thank Arjeh Cohen for stimulating their interest in this area.

For the convenience of the reader, we now give some standard definitions, several of which were used above. We also establish further notation.

Throughout,  $\mathcal{S}$  denotes an incidence system  $(\mathcal{P}, \mathcal{L})$ . It is said to have order  $s$  if every line has  $s + 1$  points, and order  $(s, t)$  if in addition every point lies on  $t + 1$  lines. If  $p$  and  $q$  lie on a line that contains three points, we shall write  $p * q$  for the third point on this line.

Distances between points are measured in the collinearity graph  $\Gamma$  of  $\mathcal{S}$ , so that  $d(p, q) = 1$  when  $p$  and  $q$  are distinct, collinear points. For  $p \in \mathcal{P}$ , and  $i = 1, 2, \dots$ , define  $\Gamma_i(p) = \{q \in \mathcal{P} : d(p, q) = i\}$  and  $\Gamma_{\leq i}(p) = \{q \in \mathcal{P} :$

$d(p, q) \leq i\}$ . These can be regarded as sets of points or as subgraphs of  $\Gamma$ . We also let  $\Gamma_{\frac{3}{2}}(p)$  and  $\Gamma_{\frac{5}{2}}(p)$  denote respectively the collection of lines containing points in both  $\Gamma_1(p)$  and  $\Gamma_2(p)$  and the collection of lines containing points in both  $\Gamma_2(p)$  and  $\Gamma_3(p)$ . The diameter of  $\mathcal{S}$  (where defined) is the smallest number  $d$  such that  $\Gamma_d(p) = \mathcal{P}$  for every point  $p$ .

In  $\mathcal{S}$ , a circuit of length  $m$  is a sequence  $p_0, p_1, p_2, \dots, p_m = p_0$  of  $m$  points such that  $p_i$  and  $p_{i-1}$  are collinear,  $i = 1, 2, \dots, m$ , but  $p_{i-1}, p_i, p_{i+1}$  do not lie on a single line. If  $\mathcal{S}$  is an incidence system of diameter  $n$  in which, for each point-line pair  $(p, L)$ , there is a unique point on  $L$  that is closest to  $p$ , then  $\mathcal{S}$  is called a near  $2n$ -gon. In that case the sets  $S(p) = \Gamma_{\leq n-1}(p)$  are geometric hyperplanes which we call special.

A generalized  $2n$ -gon is a near  $2n$ -gon  $\mathcal{S}$  in which every pair of points is contained in a circuit of length  $2n$  and there are no shorter circuits. In this case the dual incidence system  $\mathcal{S}^*$ , in which the roles of points and lines are interchanged, is also a generalized  $2n$ -gon. When  $n = 3$ , we say simply that  $\mathcal{S}$  is a hexagon.

The hexagons  $\mathcal{H}$  and  $\mathcal{H}^*$  satisfy Ronan's 'regulus condition': for any points  $p$  and  $q$  at distance 3, there is a unique point  $r$ , other than  $p$  and  $q$ , such that  $\Gamma_{\frac{3}{2}}(r) \subseteq \Gamma_{\frac{3}{2}}(p) \cap \Gamma_{\frac{3}{2}}(q)$ . We shall let  $p\#q$  denote this point. Note also that  $p\#r = q$  and  $q\#r = p$ . Sets of the form  $\{p, q, p\#q\}$  are called regulus triples.

## 2. Invariant forms and products

In this section we prove results about forms and products that will be used in the study of  $V(\mathcal{H})$  and  $V(\mathcal{H}^*)$ . These results are somewhat more general than those in the rest of this paper. It is well-known that the first one, stated as Lemma 2.1(a), holds for any incidence system  $\mathcal{S} = (\mathcal{P}, \mathcal{L})$  of order 2 (with no change in the proof below). However we shall always assume that  $\mathcal{S}$  is a near  $2n$ -gon of order 2, since all other results use that assumption. At a later stage we shall also assume that  $\mathcal{S}$  is finite. We use the notation  $U = U(\mathcal{S})$ ,  $V = V(\mathcal{S})$ , and  $G = \text{Aut}(\mathcal{S})$ .

Recall that  $U$  is generated by vectors  $v_p$ ,  $p \in \mathcal{P}$ . This gives a map  $\phi : \mathcal{P} \rightarrow U$  whose image spans  $U$ . For notational convenience we assume that  $\phi$  is injective and identify each point  $p$  with its image under  $\phi$ , but with appropriate changes in notation, that assumption is not necessary.

Given an arbitrary geometric hyperplane  $H$  (an element of  $V$ ), define  $f_H : \mathcal{P} \rightarrow \mathbb{F}_2$  by

$$f_H(p) = \begin{cases} 0 & p \in H; \\ 1 & \text{otherwise.} \end{cases}$$

If  $\{p, q, r\}$  is a line in  $\mathcal{S}$  then  $f_H(p) + f_H(q) + f_H(r) = 0$ . Therefore  $f_H$  extends to a linear functional on  $U$ , and this extension is unique because  $\mathcal{P}$  spans  $U$ . We have  $H = \phi^{-1}(\text{Ker } f_H)$ . On the other hand, if  $\alpha \in U^*$ ,

then  $\text{Ker } \alpha$  is a hyperplane of  $U$  or  $U$  itself, and  $\phi^{-1}(\text{Ker } \alpha)$  is a geometric hyperplane of  $\mathcal{S}$ . This shows that there is a natural bijection between  $V$  and  $U^*$ , the dual space of  $U$ .

We now define the sum of two geometric hyperplanes of  $\mathcal{S}$ :

$$A + B = (A \cap B) \cup (\mathcal{P} \setminus (A \cup B)), \quad A, B \in V.$$

With this operation,  $V$  can be regarded as a vector space over  $\mathbb{F}_2$ , with zero element  $0 = \mathcal{P}$ . In fact, the correspondence between linear functionals on  $U$  and geometric hyperplanes of  $\mathcal{S}$  makes it clear that  $V$  is closed under  $+$ , with this operation corresponding to addition in  $U^*$ . Elements of  $V$  are subsets of  $\mathcal{P}$ , which  $G$  permutes. This defines a  $G$ -module action on  $V$ . The first part of the following lemma is now almost immediate.

**Lemma 2.1** *Let  $\mathcal{S}$  be a near  $2n$ -gon of order 2, with  $U$ ,  $V$ , and  $G$  as above.*

(a) *The correspondence  $H \mapsto f_H$  is an  $\mathbb{F}_2 G$ -module isomorphism between  $V$  and  $U^*$ .*

(b) *The assignment  $p \mapsto S(p)$ ,  $p \in \mathcal{P}$ , induces a nonzero  $\mathbb{F}_2 G$ -module homomorphism from  $U$  to  $V$ .*

(c) *There is a symplectic form  $U \times U \rightarrow \mathbb{F}_2$  satisfying, for all  $p, q \in \mathcal{P}$*

$$(p, q) = \begin{cases} 0 & q \in S(p); \\ 1 & \text{otherwise.} \end{cases}$$

(d) *There is a quadratic form  $Q_U$  on  $U$ , with associated symplectic form as in (c), such that  $Q_U(p) = 0$  for all  $p \in \mathcal{P}$ .*

**Proof** Let  $\hat{U}$  be an  $\mathbb{F}_2$ -vector space with basis  $\mathcal{P}$  and the obvious  $G$ -action. There are evidently unique maps and forms on  $\hat{U}$  whose properties are analogous to the ones on  $U$  whose existence is asserted in (b), (c), and (d). (The bilinear form on  $\hat{U}$  is symplectic because it is symmetric and  $(p, p) = 0$  for all  $p$  in the basis  $\mathcal{P}$  of  $\hat{U}$ .) It need only be shown that these maps and forms induce corresponding ones on  $U$  via the natural linear map  $\hat{U} \rightarrow U$  whose kernel  $U_0$  is the  $G$ -submodule  $\langle p + q + r : \{p, q, r\} \in \mathcal{L} \rangle$ .

Let  $\{p, q, r\}$  be a line in  $\mathcal{S}$ . The assumptions on  $\mathcal{S}$  imply that every element of  $\mathcal{P}$  belongs to one or all of the sets  $S(p)$ ,  $S(q)$ ,  $S(r)$ . Using the addition on  $V$ , we have  $S(p) + S(q) + S(r) = \mathcal{P}$ . The linear map  $\hat{U} \rightarrow V$  taking  $p$  to  $S(p)$ ,  $p \in \mathcal{P}$ , is  $G$ -linear and contains  $U_0$  in its kernel. This induces the map in (b). Since  $\mathcal{S}$  has diameter  $n$ , some  $S(p)$  differs from 0.

For (c), we can use  $\hat{U}$  again, or we can compose the maps in (b) and (a) to produce a map  $U \rightarrow U^*$ ,  $p \mapsto f_{S(p)}$ . This gives rise to a bilinear form on  $U$  which satisfies the formula in (c). It is induced from the corresponding symplectic form on  $\hat{U}$ , whose radical must then contain  $U_0$ . A straightforward

calculation shows that  $U_0$  also lies in the radical of the quadratic form on  $\hat{U}$ , giving (d).  $\square$

The forms in (c) and (d) are clearly  $G$ -invariant. The same will be true for the forms and algebra operations (bilinear binary operations) defined below. However the remaining results apply only to finite incidence systems.

**Lemma 2.2** *Assume that  $S$  is a finite near  $2n$ -gon of order 2. Then the following hold, where  $H, H' \in V$  and  $u, v \in U$ .*

(a) *There is a bilinear form  $\langle \cdot, \cdot \rangle$  on  $V$  satisfying*

$$\langle H, H' \rangle = |\mathcal{P} \setminus (H \cup H')| \pmod{2}.$$

(b) *There is a commutative algebra operation  $[\cdot, \cdot]$  defined on  $V$  by*

$$[H, H'] = \sum_{p \notin H \cup H'} S(p).$$

(c) *There is a commutative algebra operation  $[\cdot, \cdot]$  defined on  $U$  by*

$$[u, v] = \sum_{p \in \mathcal{P}} (u, p)(v, p)p.$$

(d) *If every point of  $S$  lies on an odd number of lines then*

$$Q_v(H) = \frac{1}{2} |\mathcal{P} \setminus H| \pmod{2}$$

*is a quadratic form on  $V$  whose associated symplectic form is given by (a).*

(e) *Under the assumption of (d), any  $H \in V$  lies in the radical of the form in (a) if and only if  $|H \cap H'|$  is odd for all  $H' \in V$ .*

(f) *Under the assumption of (d), if  $V'$  is an arbitrary subspace of  $V$  on which  $\langle \cdot, \cdot \rangle$  vanishes, then the collection of all  $H$  in  $V'$  with  $|\mathcal{P} \setminus H| \equiv 0 \pmod{4}$  is subspace of  $V'$  of codimension at most 1.*

**Proof**

(a) On  $U^*$ , the form  $\langle \lambda, \mu \rangle = \sum_{p \in \mathcal{P}} \lambda(p)\mu(p)$  is bilinear. By transporting this form to  $V$ , using Lemma 2.1(a), we obtain the formula in (a).

(b) Because  $U$  and  $V$  have finite dimension, Lemma 2.1(a) also produces an isomorphism  $V^* \cong U$ . The trilinear form  $(\lambda, \mu, \nu) \mapsto \sum_{p \in \mathcal{P}} \lambda(p)\mu(p)\nu(p)$  on  $U^*$  corresponds to an element of  $\text{Hom}_{\mathbf{F}_2 G}(V \otimes V \otimes V, \mathbf{F}_2)$ , and hence to an element of  $\text{Hom}_{\mathbf{F}_2 G}(V \otimes V, U)$ . Composing this with the map  $U \rightarrow V$  from Lemma 2.1(b), we obtain an algebra (bilinear binary operation) on  $V$ . It is a straightforward matter to verify the formula given.

(c) This is obvious because the given operation is bilinear. It can also be constructed from the element of  $\text{Hom}_{\mathbb{F}_2 G}(V \otimes V, U)$  that was used in (b).

(d) Assume that the number of lines through any point in  $\mathcal{S}$  is odd. Let  $H \in V$ . Then every line of  $\mathcal{S}$  not contained in  $H$  intersects  $\mathcal{P} \setminus H$  in exactly 2 points, and there is a graph on  $\mathcal{P} \setminus H$  associated with  $H$ . A count modulo 2 of the incident point-line pairs of  $\mathcal{S}$  shows that  $|\mathcal{P}|$  is odd, and a similar count using the graph shows that  $|\mathcal{P} \setminus H|$  is even. Thus  $|H|$  is odd, and the form  $\langle \cdot, \cdot \rangle$  is symplectic.

Now let  $H_1, H_2 \in V$  and set  $H_3 = H_1 + H_2$ . Each pair of these geometric hyperplanes intersects in the same set  $I$ , which has odd cardinality if and only if  $\langle H_1, H_2 \rangle = 0$ . By considering the partition of  $\mathcal{P} \setminus I$  formed by the sets  $H_j \setminus I$ ,  $j = 1, 2, 3$  (which all have the same parity), it can be seen that

$$\frac{1}{2} \sum_j |\mathcal{P} \setminus H_j| = \sum_j |H_j \setminus I| \equiv \langle H_1, H_2 \rangle \pmod{2}.$$

This proves (d). The same calculations can be used to establish (e). Finally, (f) follows because  $Q_v|_{V'}$  is a linear functional.  $\square$

**Proposition 2.3** *Let  $\mathcal{S}$  be a generalized  $2n$ -gon of order  $(2, 2k)$ , where  $n$  and  $k$  are positive integers. Then, for all  $p \in \mathcal{P}$ ,  $S(p)$  lies in the radical of the bilinear form on  $V$  that was defined in Lemma 2.2(a).*

**Proof** If  $H \in V$  and  $\mathcal{L}_H$  consists of those lines of  $\mathcal{S}$  that are contained in  $H$ , then  $\mathcal{S}_H = (H, \mathcal{L}_H)$  is an incidence system with 3 points per line. For every  $H' \in V$ ,  $H \cap H'$  is a geometric hyperplane of  $\mathcal{S}_H$ . If  $H = S(p)$  then each point of  $H$  lies on 1 or  $2k + 1$  lines of  $\mathcal{S}_H$ . The conclusion follows from Lemma 2.2(e) and the lemma below, applied to  $\mathcal{S}_H$ .  $\square$

**Lemma 2.4** *Suppose  $\mathcal{S}'$  is a finite connected incidence system with no circuits. If every line has an odd number of points and every point lies on an odd number of lines then every geometric hyperplane of  $\mathcal{S}'$  has odd cardinality.*

**Proof** Induct on the number of points of  $\mathcal{S}'$ . Assume  $\mathcal{S}'$  has a point  $p$  on more than one line, for if this fails the assertion is trivial. Then  $\mathcal{S}'$  can be decomposed into the union of an odd number of proper subsystems satisfying the hypotheses, each pair of which intersects at  $\{p\}$ . The assertion is now easily established for any geometric hyperplane whether or not it contains  $p$ .  $\square$

**Remark** The previous lemma can also be applied to certain geometric hyperplanes which have an odd number of connected components.

### 3. The universal embedding spaces of the hexagons of order $(2, 2)$

In this section, let  $\mathcal{S}$  be either  $\mathcal{H}$  or  $\mathcal{H}^*$ , and set  $U = U(\mathcal{S})$ , the universal module for  $\mathcal{S}$ . A simple ‘duality’ argument (sketched below in Corollary 3.3) shows that the two universal spaces have the same dimension. Computer calculations of S. Smith showed that this dimension is 14. A conceptual proof, with the technical details omitted, was sketched by Ronan [9]. It is also a particular case of a far more general theorem of Völklein [10]. Here we give a short new proof of this fact and obtain related results that are applied in [8] for spanning arguments in larger geometries.

We first describe some properties common to both hexagons of order  $(2, 2)$ . Here we rely heavily on the analyses of Cohen and Tits [2]. Fix a point  $p$  of  $\mathcal{S}$ , and let  $L_1, L_2, L_3$  be the lines containing  $p$ . Also, let  $M_i$  be the set of points in  $\Gamma_2(p)$  lying at distance 1 from some point in  $L_i$ ,  $i = 1, 2, 3$ .

We shall often consider the subgraphs  $\Gamma_i = \Gamma_i(p)$  of the incidence graph  $\Gamma$  of  $\mathcal{S}$ . In particular,  $\Gamma_3$  is useful for studying spanning properties in  $U(\mathcal{S})$ . Its vertices are the points at maximal distance from  $p$ . This graph contains 32 points and 48 edges. Each of these edges comes from an line of  $\mathcal{S}$  containing a unique point of  $\Gamma_2$  in addition to the two points in  $\Gamma_3$ . The mapping so defined, from the edges of  $\Gamma_3$  to the 24 points of  $\Gamma_2$ , is 2 to 1.  $\Gamma_3$  is connected when  $\mathcal{S} = \mathcal{H}$  and has two components when  $\mathcal{S} = \mathcal{H}^*$ . In the latter case the mapping restricts to a one-to-one correspondence between the edges of either component and  $\Gamma_2$ . For convenience, we shall occasionally refer to the points of  $\Gamma_2$  as edges, even though each point occurs twice as an edge in the full  $\Gamma_3$  graph.

Suppose  $x_0, x_1, \dots, x_k = x_0$  is a circuit in  $\Gamma_3$ . Then  $y_i = x_{i-1} * x_i \in \Gamma_2$ ,  $i = 1, \dots, k$ . Say that such a circuit has type  $j_1 j_2 \dots j_k$  when  $y_i \in M_{j_i}$ ,  $i = 1, \dots, k$ . Minimal circuits in  $\Gamma_3$  have length 6. As shown in [2], such circuits have type  $abcabc$  in  $\mathcal{H}$  and (by choosing the appropriate starting point) type  $abcacb$  in  $\mathcal{H}^*$ , where  $\{a, b, c\} = \{1, 2, 3\}$ . If  $k = 8$  and the circuit has type  $abababab$  we say that the set  $\{y_1, \dots, y_8\} \subseteq \Gamma_2$  is an octagon of type  $ab$ . In this situation  $y_i$  and  $y_{i+4}$  are collinear and distinct,  $i = 1, \dots, 4$ , as are  $y_i * y_{i+4}$  and  $y_{i+2} * y_{i+6}$ ,  $i = 1, 2$ . Note that in the terminology of graphs introduced above the points  $y_i$  are actually ‘edges.’ It can be shown that they uniquely determine the octagon, whose vertices form a circuit of length 8 in  $\Gamma_3$ .

The following spanning result is particularly useful when each  $x_i$  belongs to  $\Gamma_3$ . All sums are taken within  $U$ .

**Lemma 3.1** *If  $x_0, x_1, \dots, x_k = x_0$  is a circuit in  $\mathcal{S}$  and  $y_i = x_{i-1} * x_i$ ,  $i = 1, \dots, k$ , then  $\sum_{i=1}^k y_i = 0$ . In particular, every subspace of  $U$  containing at least  $k - 1$  of the points  $y_i$  contains all  $k$  points  $y_i$ .*

**Proof** For  $i = 1, \dots, k$ , we have  $y_i = x_{i-1} + x_i$ . Therefore  $\sum_{i=1}^k y_i = x_0 + 2 \sum_{i=1}^{k-1} x_i + x_k = 0$ . The second assertion is an immediate consequence.  $\square$

For the remainder of this section let  $U = U(\mathcal{H}^*)$ . There is a well-known embedding of  $\mathcal{H}^*$  into  $\mathfrak{g}_2(2)$  (or any adjoint  $G_2(2)$ -module) which is described in detail in [7]. The points of  $\mathcal{H}^*$  span the whole space. Thus  $\dim(U) \geq 14$ . We shall soon see a new proof that equality holds — in other words, the adjoint embedding is universal. Henceforth we identify  $\mathcal{H}^*$  with its image in  $U$ .

**Proposition 3.2** *Assume that  $W$  is a subspace of  $U(\mathcal{H}^*)$  such that  $L_1 \cup L_2 \cup M_1 \cup M_2 \subseteq W$ . Then  $L_3 \subseteq W$ . If in addition  $W$  contains a point of  $M_3$ , then  $M_3 \subseteq W$ .*

**Proof** Let  $\Delta$  be an arbitrary connected component of  $\Gamma_3$ . It is the disjoint union of two octagons of type 12. If  $q$  and  $q'$  are adjacent vertices of one of these octagons then  $q * q' \in W$ , and hence  $q + W = q' + W$  in the quotient space  $U/W$ . Of course the other octagon also determines a unique element of  $U/W$ . Let  $r$  denote an arbitrary element of  $M_3$ . It lies on a line containing a point from each octagon, so  $r + W$  does not depend on the choice of  $r$ . Finally, let  $s$  denote either point in  $L_3 \cap \Gamma_1$ . Since there is a line through  $s$  whose other two points lie in  $M_3$ , we have  $s + W = r + r + W$ , so  $s \in W$ . Therefore  $L_3 \subseteq W$ . The last assertion of the Proposition is also clear.  $\square$

**Corollary 3.3** *The universal embedding space for  $\mathcal{H}^*$  is 14-dimensional.*

**Proof** We shall choose fourteen points in  $U$ , then prove that they span  $U$ . Start with any two points on  $L_1$  and any point except  $p$  on  $L_2$ . Note that, for  $i = 1, 2, 3$ ,  $M_i$  consists of the points in  $\Gamma_2$  lying on the four lines in  $\Gamma_{\frac{3}{2}} = \Gamma_{\frac{3}{2}}(p)$  that intersect  $L_i$ . Thus we can choose eight more points in  $M_1 \cup M_2$ , no two collinear. Choose the twelfth point to be in  $M_3$ , then one more in each of the two connected components of  $\Gamma_3$ .

Let  $W$  denote the span of the points just chosen. Since  $\dim(W) \leq 14$ , we need only show that  $W = U$ . It follows in succession that  $L_1 \subseteq W$ ,  $L_2 \subseteq W$ , and  $M_1 \cup M_2 \subseteq W$ . By our previous result,  $L_3 \subseteq W$  and also  $M_3 \subseteq W$ . It follows rapidly that every point of  $\mathcal{H}^*$  lies in  $W$  and hence  $W = U$ .  $\square$

**Corollary 3.4** *The universal embedding space for  $\mathcal{H}$  is 14-dimensional.*

This follows from an elementary and well-known ‘duality’ argument, based on the fact that  $\mathcal{H}$  and  $\mathcal{H}^*$  have  $63 \times 63$  incidence matrices which are transposes of each other and therefore have the same rank  $k$  over  $\mathbf{F}_2$ . Both

embedding spaces then have rank  $63 - k$ . (However, as  $G_2(2)$ -modules,  $U(\mathcal{H})$  and  $U(\mathcal{H}^*)$  have different structures.)  $\square$

We continue studying properties of the embedding of  $\mathcal{H}$  in  $U$  in order to obtain results which are used in [8] when determining the universal embedding space of the generalized hexagon of order  $(2, 8)$ .

Recall from [2] that each component of  $\Gamma_3$  has a labelling that makes it a 2-cover of the 3-cube. In this labelling, every point is labelled by a 3-tuple of 0s and 1s. Points that are connected by an edge of type  $i$  have labels that differ at index  $i$  and agree at the other indices. We observe that if  $O$  is an octagon of type 12, say, with associated circuit  $C$ , and  $\Delta$  is a component of  $\Gamma_3$  containing  $C$  then all points in  $C$  have the same 3-label, and in fact  $C$  consists of all points in  $\Delta$  having that 3-label.

**Proposition 3.5** *If  $W$  is a subspace of  $U$  containing  $\Gamma_1 \cup M_1$  then  $\Gamma_2 \subseteq W$ , provided  $W$  satisfies one of the following further conditions.*

- (a)  *$W$  contains three points in  $M_2$ , no two collinear, and a point in  $M_3$ .*
- (b)  *$W$  contains four points of a 23-octagon.*

**Proof** (a) Let  $\{q, q', q''\}$  be a line with  $q \in L_2$ , and  $q', q'' \in M_2$ . Then  $q' \in W$  exactly when  $q'' \in W$ . Thus (a) implies that three of the four lines joining points of  $L_2$  to points of  $M_2$  must be contained in  $W$ . We establish that  $M_2 \subseteq W$  by using Lemma 3.1 to show that  $W$  contains a point on the fourth line. Choose a component  $\Delta$  of  $\Gamma_3$  and a point  $r \in M_3 \cap W$ . Then  $r \in \Gamma_2$  corresponds to a unique edge of  $\Delta$  with vertices, say,  $a$  and  $b$ . Let  $a' \neq a$ ,  $b' \neq b$  be the points of  $\Delta$  that share the labels of  $a$ ,  $b$ , and let  $r' = a' * b'$ . Then  $r$  and  $r'$  are collinear and  $r * r' \in \Gamma_1$ . We therefore have  $r' \in W$ . The point  $a$  is contained in a unique circuit of type 12121212, and the eight edges in this circuit are points in  $\Gamma_2$  lying on four lines  $N_1, \dots, N_4$ , of  $\Gamma_{\frac{3}{2}}$ , two of which meet  $L_1$ , and two of which meet  $L_2$ . On this circuit,  $a'$  lies at distance 4 from  $a$ . If  $y_1, \dots, y_4$  are the edges from one of the half-circuits joining  $a$  to  $a'$  then every  $N_i, i = 1, \dots, 4$  must contain some  $y_j$  because no pair of  $y_j$ s can be collinear. Similar considerations show that the edge set of a path of type 1212 from  $b$  to  $b'$  must meet each of the four lines  $N_5, \dots, N_8$  in  $\Gamma_{\frac{3}{2}}$ , involving edges in a circuit of type 12121212 containing  $b$ . But these four lines are precisely the lines in  $\Gamma_{\frac{3}{2}}$ , different from  $N_1, \dots, N_4$ , that meet  $L_1$  or  $L_2$  and contain points in  $\Gamma_2$ . Consider the circuit of type 1212312123 that starts at  $a'$ , goes to  $a$  along a path of length 4, then to  $b$  via  $r$ , and to  $b'$  by another path of length 4 before returning along  $r'$  to  $a'$ . All edges of type 1 belong to  $W$ , as do  $r$  and  $r'$ , the two edges of type 3. Since  $W$  contains at least three of the four lines joining points in  $M_2$  to  $L_2$ , it follows that  $W$  contains all but at most one of the edges in this circuit, whence all of the

edges in the circuit, by Lemma 3.1. This implies that every line joining a point of  $M_2$  to a point of  $L_2$  must meet  $M_2 \cap W$  and that  $M_2 \subseteq W$ .

Now let  $r \in M_3 \cap W$  and let  $a \in M_3$ . Choose a component  $\Delta$  of  $\Gamma_3$ . Then  $\Delta$  contains a circuit with edges  $a, r$ , and points in  $M_1 \cup M_2$ . Since  $\{a\} \cup M_1 \cup M_2 \subseteq W$ , Lemma 3.1 implies that  $a \in W$ .

(b) In this case the remarks above show that  $W$  contains an octagon  $O$  of type 23. We need to prove that  $M_2 \cup M_3 \setminus O \subseteq W$ . Let  $q \in M_2 \cup M_3 \setminus O$ , and let  $\Delta$  be the component of  $\Gamma_3$  containing the circuit associated to  $O$ . Then  $q$  is an edge in  $\Delta$  whose endpoints are connected to the circuit of  $O$  by edges of type 1. There is therefore a circuit in  $\Delta$  where the edges consist of  $q$ , two elements of  $M_1$ , and a subset of  $O$ . The hypotheses and Lemma 3.1 now imply that  $q \in W$ .  $\square$

#### 4. Geometric hyperplanes of $\mathcal{H}$ and the module structure of $V(\mathcal{H})$

The classical hexagon (over any field) is usually described in terms of its embedding in a 7-dimensional vector space  $W$ . This hexagon consists of the points and some of the lines of an orthogonal geometry, defined from a quadratic form  $Q_w$ . For each point  $p$  of the hexagon, the hyperplane  $p^\perp$  of  $W$ , computed using the associated symmetric bilinear form, intersects the embedded hexagon at the special geometric hyperplane  $S(p) = \Gamma_{\leq 2}(p)$ . Over a field of characteristic 2, the bilinear form is symplectic, with 1-dimensional radical. If the field is also finite (or perfect), there is a natural bijection between the  $S(p)$ s and the hyperplanes of  $W$  containing the radical.

In the case of interest here, we are working over  $\mathbf{F}_2$ , with  $U = U(\mathcal{H})$  and  $G = G_2(2)$ . The universal property of  $U$  gives a natural map from  $U$  to  $W$  which is evidently  $G$ -linear. Also, the quadratic form on  $\hat{U}$  has a universal property which, using the above description of the embedding in  $W$ , leads to a homomorphism  $(U, Q_U) \rightarrow (W, Q_w)$  of orthogonal spaces. It follows that  $Q_U$  is the natural pullback (with large radical) of the quadratic form  $Q_w$  most usually associated with  $\mathcal{H}$ .

It is very convenient to work within  $V = V(\mathcal{H})$ , the space of geometric hyperplanes of  $\mathcal{H}$ , because of its concrete nature. This is a  $G$ -module which, by Lemma 2.1(a), is dual to  $U$ . As usual,  $V^\#$  denotes  $V$  with the zero vector 0 (namely  $\mathcal{P}$ ) removed.

We can associate a 5-tuple  $(n; n_0, n_1, n_2, n_3)$  with any geometric hyperplane  $H$ , where  $n = |H|$  and  $n_i$  is the number of points in  $H$  that belong to exactly  $i$  lines contained in  $H$ . The elements of this 5-tuple will be called the parameters of  $H$ . It is clear that two geometric hyperplanes in the same  $G$ -orbit have the same parameters.

The  $G$ -orbits  $\mathcal{V}_i$  ( $1 \leq i \leq 25$ ) on  $V^\sharp$  are listed by decreasing order of stabilizers, grouping together orbits whose representatives are stabilized by the same conjugacy class of subgroups of  $G$ . We also include the dimension of the  $G$ -submodule generated by each orbit. It turns out that there is only one pair of distinct  $G$ -orbits (namely  $\mathcal{V}_{24}$  and  $\mathcal{V}_{25}$ ) which cannot be distinguished by the values of their parameters alone. A few types of geometric hyperplanes were described by Ronan [9].

The information in Table 1 was obtained essentially as follows. We constructed geometric hyperplanes in three ways: directly from the geometry of  $\mathcal{H}$ , from the action of various subgroups on  $\mathcal{H}$ , as outlined in the following paragraph, and by forming sums within  $V$ . Since  $\mathcal{H}$  (and also  $\mathcal{H}^*$ ) is rather small, with only 63 points and lines, diagrams of the form given in [2] were easy to use. After finding a geometric hyperplane, we determined its  $G$ -stabilizer by hand and checked the results with the CAYLEY system. This process continued until the sum of the sizes of the known orbits (including the trivial orbit  $\{\mathcal{P}\}$ ) reached  $2^{14}$ . The proof of Proposition 4.1 shows how to determine the dimensions of the submodule generated by each  $\mathcal{V}_i$ .

For a given  $K \leq G$ , the  $K$ -invariant geometric hyperplanes can be constructed from a sort of incidence relation between the  $K$ -orbits  $\mathcal{P}/K$  on  $\mathcal{P}$  and the  $K$ -orbits  $\mathcal{L}/K$  on  $\mathcal{L}$ , in which multiplicities are counted. This relation is particularly easy to work with when  $K$  has a small number of orbits on  $\mathcal{P}$ . If  $\mathcal{Q} \subseteq \mathcal{P}$  is  $K$ -invariant and  $L, L' \in \mathcal{L}$  lie in the same  $K$ -orbit, then  $|\mathcal{Q} \cap L| = |\mathcal{Q} \cap L'|$ . We can define the incidence number between a  $K$ -invariant subset  $\mathcal{Q}$  of  $\mathcal{P}$  and a  $K$ -orbit  $\mathcal{L}_i$  on  $\mathcal{L}$  to be  $|\mathcal{Q} \cap L|$  for some (hence every)  $L \in \mathcal{L}_i$ . The  $K$ -invariant geometric hyperplanes are exactly those unions  $\mathcal{Q}$  of  $K$ -orbits on  $\mathcal{P}$  that have incidence number 1 with every line not contained in  $\mathcal{Q}$ , or in other words, incidence number 1 or 3 with every  $K$ -orbit on  $\mathcal{L}$ . Knowing the  $K$ -orbits on  $\mathcal{P}$  and  $\mathcal{L}$  it is a straightforward matter to construct the collection of  $K$ -invariant geometric hyperplanes in this way.

We describe the results of this procedure for the Weyl group and the Sylow subgroups and of  $G$ . Note that since  $G$  is defined over  $\mathbb{F}_2$ , it has trivial Cartan subgroups. Then the Weyl group of  $G$  can be identified with a subgroup  $N \cong D_{12}$  forming part of a  $B$ - $N$  pair of  $G$ . It can be shown that  $C_V(N)$  has dimension 4, and its fifteen nonzero points lie in 14 different  $G$ -orbits, namely  $\mathcal{V}_i$ ,  $1 \leq i \leq 6$ ,  $i = 8, 9$ , or  $17 \leq i \leq 22$ .

Let  $S \in \text{Syl}_7(G)$ . Then  $C_V(S)$  has dimension 2 and contains three distinct nonzero geometric hyperplanes belonging to  $\mathcal{V}_2$ ,  $\mathcal{V}_3$ , and  $\mathcal{V}_4$  in the notation of Table 1. Each of these geometric hyperplanes has the same stabilizer, a group isomorphic to  $\text{PGL}_2(7)$ , which has index 36 in  $G$ . Therefore  $|\mathcal{V}_2| = |\mathcal{V}_3| = |\mathcal{V}_4| = 36$ .

**TABLE 1:** *G*-orbit representatives of geometric hyperplanes in  $\mathcal{H}$ 

|                    | Parameters      | stabilizer              | number | $\dim(\langle \mathcal{V}_i \rangle)$ |
|--------------------|-----------------|-------------------------|--------|---------------------------------------|
| $\mathcal{V}_1$    | (27;0,27,0,0)   | $X_{27}^+ : QD_{16}$    | 28     | 7                                     |
| $\mathcal{V}_2$    | (21;21,0,0,0)   | $PGL(2, 7)$             | 36     | 14                                    |
| $\mathcal{V}_3$    | (35;0,21,0,14)  | $PGL(2, 7)$             | 36     | 7                                     |
| $\mathcal{V}_4$    | (49;0,0,21,28)  | $PGL(2, 7)$             | 36     | 14                                    |
| $\mathcal{V}_5$    | (45;0,0,27,18)  | $X_{27}^+ : D_8$        | 56     | 14                                    |
| $\mathcal{V}_6$    | (31;0,24,0,7)   | $(4 \times 4) : D_{12}$ | 63     | 6                                     |
| $\mathcal{V}_7$    | (23;16,6,0,1)   | $(4 \times 4) : S_3$    | 126    | 13                                    |
| $\mathcal{V}_8$    | (27;8,15,0,4)   | $2 \times S_4$          | 252    | 13                                    |
| $\mathcal{V}_9$    | (43;0,3,24,16)  | $2 \times S_4$          | 252    | 13                                    |
| $\mathcal{V}_{10}$ | (39;0,10,16,13) | $8 : 2 : 2$             | 378    | 13                                    |
| $\mathcal{V}_{11}$ | (25;10,12,3,0)  | $S_4$                   | 504    | 14                                    |
| $\mathcal{V}_{12}$ | (29;7,12,6,4)   | $S_4$                   | 504    | 14                                    |
| $\mathcal{V}_{13}$ | (27;8,11,8,0)   | $D_{16}$                | 756    | 13                                    |
| $\mathcal{V}_{14}$ | (33;4,8,17,4)   | $D_{16}$                | 756    | 14                                    |
| $\mathcal{V}_{15}$ | (37;1,8,20,8)   | $D_{16}$                | 756    | 14                                    |
| $\mathcal{V}_{16}$ | (35;0,13,16,6)  | $D_{16}$                | 756    | 13                                    |
| $\mathcal{V}_{17}$ | (27;6,15,6,0)   | $N \cong D_{12}$        | 1008   | 13                                    |
| $\mathcal{V}_{18}$ | (29;5,12,12,0)  | $N$                     | 1008   | 14                                    |
| $\mathcal{V}_{19}$ | (29;6,12,9,2)   | $N$                     | 1008   | 14                                    |
| $\mathcal{V}_{20}$ | (33;2,12,15,4)  | $N$                     | 1008   | 14                                    |
| $\mathcal{V}_{21}$ | (35;2,9,18,6)   | $N$                     | 1008   | 13                                    |
| $\mathcal{V}_{22}$ | (37;0,12,15,10) | $N$                     | 1008   | 14                                    |
| $\mathcal{V}_{23}$ | (29;4,16,7,2)   | $D_8$                   | 1512   | 14                                    |
| $\mathcal{V}_{24}$ | (31;4,12,12,3)  | $D_8$                   | 1512   | 13                                    |
| $\mathcal{V}_{25}$ | (31;4,12,12,3)  | $S_3$                   | 2016   | 13                                    |

If  $T \in \text{Syl}_3(G)$ , then  $C_V(T)$  again has dimension 2 and contains three nonzero geometric hyperplanes. One of these belongs to  $\mathcal{V}_1$ , and the other two belong to  $\mathcal{V}_5$ . The latter two are interchanged by  $N(T)$ , which is the unique maximal subgroup of  $G$  containing  $T$  and distinct from  $G'$ . This shows that  $|\mathcal{V}_1| = 28$  and  $|\mathcal{V}_5| = 56$ .

A Sylow 2-subgroup  $R$  of  $G$  fixes a unique nonzero geometric hyperplane, namely  $S(p) = \Gamma_{\leq 2}(p)$ , where  $p$  is the unique fixed point of  $R$  on  $\mathcal{P}$ . We have  $S(p) \in \mathcal{V}_6$ . The stabilizer of  $S(p)$  is  $G_p$ , a maximal parabolic subgroup of  $G$  having index 63, so  $|\mathcal{V}_6| = 63$ .

Several kinds of geometric hyperplanes of  $\mathcal{H}$  have easy geometric descriptions. We have already discussed the ones in  $\mathcal{V}_6$ . The elements of  $\mathcal{V}_1$  consist of the union of 9 disjoint lines in  $\mathcal{H}$ , while the elements of  $\mathcal{V}_2$  are sometimes called 'ovoids' since each of them contains exactly one point from every line.

For each point  $p$ , there are two elements of  $\mathcal{V}_7$  having stabilizers with index 2 in  $G_p$ , which can be constructed as follows. The graph  $\Gamma_3 = \Gamma_3(p)$  is a connected bipartite graph by [2], and we can select (in exactly two ways) a subset  $\Delta \subseteq \Gamma_3(p)$  of 16 points such that every line in  $\Gamma_{\frac{5}{2}}(p)$  contains a unique point in  $\Delta$ . It is easy to see that  $H_0 = \Gamma_{\leq 1}(p) \cup \Delta$  is a geometric hyperplane and  $H_0 \in \mathcal{V}_7$ .

#### Proposition 4.1

- (a)  $V$  is a uniserial  $\mathbf{F}_2 G$ -module of type  $1/6/1/6$ .
- (b)  $U$  is a uniserial  $\mathbf{F}_2 G$ -module of type  $6/1/6/1$ .

**Proof** Because  $U$  is dual to  $V$ , we need only prove (a). We first argue that  $V$  has a submodule  $V_6$  of dimension 6, a submodule  $V_7$  of dimension 7, and that  $V_6$  is the unique minimal submodule of  $V$ . If  $V'$  is a non-trivial submodule of  $V$  and  $R \in \text{Syl}_2(G)$ , then  $C_{V'}(R) \neq 0$ , so  $V'$  must contain an element of  $\mathcal{V}_6$ . In fact,  $\mathcal{V}_6 \subseteq V'$ . That the sum of two distinct elements of  $\mathcal{V}_6$  belongs to  $\mathcal{V}_6$  is a straightforward calculation involving three cases. This shows that  $V_6 = \mathcal{V}_6 \cup \{\mathcal{P}\}$  is the unique minimal submodule of  $V$ . The classical embedding of  $\mathcal{H}^*$  in a 7-dimensional vector space  $W$  gives rise to a submodule  $V_7$  of  $V$  consisting of the geometric hyperplanes obtained in the usual way from hyperplanes of  $W$ . A short calculation involving a small number of cases shows that  $V_7 = V_6 \cup \mathcal{V}_1 \cup \mathcal{V}_3$ . Here  $V_6$  arises from a 6-dimensional quotient of  $W$ . Further geometric details appear in [9].

We next argue that  $V$  has no submodule of dimension 8. Suppose  $V_8$  is such a submodule. We have seen that a Sylow 7-subgroup  $S$  of  $G$  has a 2-dimensional fixed-point submodule. Since  $N(S)$  is a Frobenius group of order 42, it follows that the irreducible  $N(S)$  constituents of  $V$  have dimensions 1

and 6, with two of each dimension. Thus  $V_8$  must contain  $C_V(S)$ , and hence  $\mathcal{V}_2, \mathcal{V}_3$ , and  $\mathcal{V}_4$ . In particular,  $V_8$  contains a geometric hyperplane  $H \in \mathcal{V}_2$  consisting of 21 points, no two of which are collinear. Let  $p \in H$ . Then  $S(p) \in V_6 \subseteq V_8$ , so  $K = S(p) + H \in V_8$ . Since  $S(p) \cap H$  consists of  $p$  itself and one point on each of the lines in  $\Gamma_{\frac{3}{2}}(p)$ , it follows that  $|K| = 37$ . In fact  $K \in \mathcal{V}_{15}$ , so  $\mathcal{V}_{15} \subseteq V_8$ . But this is impossible because  $|\mathcal{V}_{15}| = 756 > 2^8$ . Therefore  $V$  has no submodule of dimension 8.

This implies that  $V_7$  is the unique submodule of dimension 7. If  $V'$  is a non-trivial submodule of  $V$  different from  $V_6$  and  $V_7$ , then  $\dim V' > 8$ . The consideration of the  $N(S)$  action shows that if  $V'$  is proper then either  $\dim V' = 12 = 6 + 6$  or  $\dim V' = 13 = 6 + 6 + 1$ .

Suppose  $V_{12}$  is a submodule of dimension 12. Then  $N(S)$ , and hence  $G$ , acts trivially on  $V/V_{12}$ . Since  $T \in \text{Syl}_3(G)$ ,  $V$  is completely reducible as a  $T$ -module. We have  $V = V_{12} \oplus X$ , where  $X \subseteq C_V(T)$ . In fact  $X = C_V(T)$ , by the geometric method described above (or by considering the action of  $T$  on  $V_{12}$  and its submodule  $V_6$ ). However we have seen that  $N(T)$  acts nontrivially on  $C_V(T)$ , by interchanging two members of  $\mathcal{V}_5$ . But then  $G$  acts nontrivially on  $V/V_{12}$ . This contradiction shows that  $V$  has no submodule  $V_{12}$ .

The non-existence of a 12-dimensional submodule implies that there is at most one submodule  $V_{13}$  of dimension 13. That one exists follows easily from the 2-modular theory of  $G$  (which excludes a 7-dimensional irreducible module), but there is no need to invoke that here. We can describe  $V_{13}$  in terms of the geometry. Consider the form  $\langle \cdot, \cdot \rangle$  on  $V$  that was defined in Lemma 2.2(a). The radical  $V_0$  of this form is a  $G$ -submodule of  $V$ . Then  $\mathcal{V}_6 \subseteq V_0$  by Proposition 2.3. The remark following Lemma 2.4 implies that  $\mathcal{V}_1 \subseteq V_0$  as well. Therefore,  $V_7 \subseteq V_0$ . Since  $\langle \cdot, \cdot \rangle$  is a symplectic form, by Lemma 2.2(d),  $V/V_0$  is even-dimensional. The only possibility not excluded by previous arguments is  $V_0 = V$ . By Lemma 2.2(f), the form  $Q_v$  defined in Lemma 2.2(d) is a linear functional. The kernel of  $Q_v$  consists of all geometric hyperplanes with co-cardinality divisible by 4. This is a proper subset of  $V$ , so it forms the desired submodule  $V_{13}$ . Moreover, it contains  $V_7$ . Thus  $V$  is a uniserial  $G$ -module.  $\square$

The theory that we have developed has a surprising consequence.

**Proposition 4.2** *In  $\mathcal{H}^*$ , any two geometric hyperplanes intersect at a set of odd cardinality.*

**Proof** We have just seen that the form  $\langle \cdot, \cdot \rangle$  on  $V$ , defined is trivial. By the definition of this form,  $|\mathcal{P} \setminus H \cup H'|$  is even, where  $H$  and  $H'$  are geometric hyperplanes of  $\mathcal{H}^*$ . Since  $|H|$ ,  $|H'|$ , and  $|\mathcal{P}|$  are odd, this implies that  $|H \cap H'|$  is also odd.  $\square$

TABLE 2:  $G$ -orbits of geometric hyperplanes in  $\mathcal{H}^*$

|                    | Parameters         | stabilizer                 | number | weight |
|--------------------|--------------------|----------------------------|--------|--------|
| $\mathcal{W}_1$    | (31; 0, 24, 0, 7)  | $X_{32}^+ : S_3$           | 63     | 1      |
| $\mathcal{W}_2$    | (47; 0, 0, 24, 23) | $(4 \circ Q_8) : S_3$      | 126    | 3      |
| $\mathcal{W}_3$    | (23; 16, 6, 0, 1)  | $Q_8 : S_3$                | 252    | 4      |
| $\mathcal{W}_4$    | (39; 0, 6, 24, 9)  | $2 \times S_4$             | 252    | 4      |
| $\mathcal{W}_5$    | (39; 0, 9, 18, 12) | $S_3 \times S_3$           | 336    | 3      |
| $\mathcal{W}_6$    | (31; 0, 20, 8, 3)  | $(2 \times 4) \cdot (2^2)$ | 378    | 2      |
| $\mathcal{W}_7$    | (35; 0, 14, 14, 7) | $D_{14}$                   | 864    | 3      |
| $\mathcal{W}_8$    | (31; 4, 15, 6, 6)  | $D_{12}$                   | 1008   | 2      |
| $\mathcal{W}_9$    | (39; 0, 9, 18, 12) | $D_{12}$                   | 1008   | 3      |
| $\mathcal{W}_{10}$ | (27; 8, 12, 6, 1)  | $S_3$                      | 2016   | 3      |
| $\mathcal{W}_{11}$ | (35; 2, 9, 18, 6)  | $S_3$                      | 2016   | 4      |
| $\mathcal{W}_{12}$ | (27; 6, 15, 6, 0)  | $S_3$                      | 2016   | 3      |
| $\mathcal{W}_{13}$ | (31; 4, 11, 14, 2) | $2^2$                      | 3024   | 4      |
| $\mathcal{W}_{14}$ | (31; 4, 12, 12, 3) | $2^2$                      | 3024   | 3      |

### 5. Geometric hyperplanes of $\mathcal{H}^*$

Let  $V = V(\mathcal{H}^*)$ ,  $U = U(\mathcal{H}^*)$ , and, as before,  $G = G_2(2)$ . As discussed in Section 3 and elsewhere,  $U$  is isomorphic to the irreducible 14-dimensional adjoint module for  $G$ . Since the map  $U \rightarrow V$  constructed in Section 2 is non-trivial, and  $V \cong U^*$ , this gives a canonical way to identify the  $G$ -modules  $U$  and  $V$ , with each point  $p$  of the embedded image of  $\mathcal{H}^*$  in  $U$  corresponding to the geometric hyperplane  $S(p)$ . It is notationally more convenient to work in  $U$ , shifting to  $V$  whenever geometric hyperplanes are being discussed.

By the irreducibility of  $V$ , every geometric hyperplane of  $\mathcal{H}^*$  is of the form  $\sum_{p \in X} S(p)$  for some finite subset  $X$  of  $\mathcal{P}$ . Define the weight of a geometric hyperplane to be the size of the smallest such  $X$ . In Table 2 we record information about the  $G$ -orbits on  $V$ . One useful consequence is that the parameters of a geometric hyperplane determine its  $G$ -orbit unless it belongs to  $\mathcal{W}_5 \cup \mathcal{W}_9$ . In the latter case, the structures of  $G$ -stabilizers distinguish the orbits. The types of geometric hyperplanes lying in various subspaces can be determined by hand, but we found it convenient to use our own C program to find the parameters, resolving ambiguous cases with Cayley. The only pair of orbits with the same conjugacy class of stabilizers is  $\{\mathcal{W}_{10}, \mathcal{W}_{11}\}$ .

**Remark** Because the special geometric hyperplanes  $S(p)$  span  $V$ , the form  $\langle \cdot, \cdot \rangle$  defined in Section 2 is trivial, by Proposition 2.3. Thus every intersection of two geometric hyperplanes of  $\mathcal{H}^*$  has odd order.

Although it is possible to find every element of  $V$  by constructing all geometric hyperplanes of weights 1, 2, 3, 4, it is easier to generate representatives for each  $G$ -orbit by adding pairs of previously constructed geometric hyperplanes, then determining the group stabilizer. A counting argument determines when all  $G$ -orbits on  $V$  have been found. It is easy to describe the geometric hyperplanes of weight 2 or less. Clearly  $S(p) \in \mathcal{W}_1$ , while  $S(p) + S(q)$  belongs to  $\mathcal{W}_6$  when  $q \in \Gamma_2(p)$  and to  $\mathcal{W}_8$  when  $q \in \Gamma_3(p)$ . (By Lemma 2.1(b),  $S(p) + S(q) = S(p * q) \in \mathcal{W}_1$  when  $q \in \Gamma_1(p)$ .)

We now proceed to describe specific representatives of the  $G$ -orbits on  $U$  within two subspaces that will be used in the next section in the discussion of the Lie algebra properties of  $U$ . For convenience, we first note the following result. The first part could have been phrased in terms of apartments of the building  $\mathcal{H}^*$ .

**Lemma 5.1**

- (a)  $G$  acts regularly on ordered circuits of length 6 in  $\mathcal{H}^*$  and transitively on paths of length 3.
- (b) If  $\{p_1, p_2, p_3, p_4\}$  is a path of length 3 in  $\mathcal{H}^*$ , then there is a unique involution  $t \in G$  such that  $t$  fixes  $\{p_1, p_2, p_3, p_4\}$  pointwise.
- (c)  $t$  fixes exactly 24 ordered paths of length 3 in  $\mathcal{H}^*$ .

**Proof** Statements (a) and (b) follow from distance transitivity together with the fact that the stabilizer of two points at distance 3 acts naturally as  $S_3$  on the three paths of minimal length connecting them. Choosing a path and an involution as in (b), it is clear that  $t$  must also fix  $p_1 * p_2$ ,  $p_2 * p_3$ ,  $p_3 * p_4$ ,  $p_1 \# p_4$ , and  $(p_2 * p_3) * (p_1 \# p_4)$ . Geometric methods show that  $t$  fixes no further points. This can also be seen by using the Atlas [3] to examine values of the permutation character of  $G$  from its action on the 63 points of  $\mathcal{H}^*$ . The nine points fixed by  $t$  contain four lines of  $\mathcal{H}^*$ , arranged in a configuration which clearly contains exactly 24 ordered paths of length 3. This proves (c).  $\square$

Using the Lie algebra, we shall show in Section 6 that there is a unique element  $v$  of  $\mathcal{W}_4$  such that  $G_v = C_G(t)$ .

Fix the following notation for the balance of the paper. Choose a circuit  $p_1, p_2, \dots, p_6$  of length 6 in  $\mathcal{H}^*$ , and let  $p'_1$  be the unique point in  $\Gamma_2(p_5)$  which is collinear with  $p_2$  but not with  $p_1$  or  $p_3$ . Let  $p_{14} = p_1 \# p_4$ ,  $p'_{14} = p'_1 \# p_4$ , and  $p'_4 = p_1 \# p'_{14}$ .

Straightforward calculations within the module of geometric hyperplanes of  $\mathcal{H}^*$  show that  $U_1 = \langle p_1, p_2, p_3, p_4, p_5, p'_1, p'_4, p_{14} \rangle$  contains representatives of every  $G$  orbit on  $U$ . We record a list of such representatives in Table 3. Half of the  $G$ -orbits can also be found in a subspace  $B$ , associated with the

TABLE 3:  $G$ -orbit representatives in  $U^\sharp$

|                    | In $U_1$                           | In $B$                              |
|--------------------|------------------------------------|-------------------------------------|
| $\mathcal{W}_1$    | $p_1$                              | $x_{00}$                            |
| $\mathcal{W}_2$    | $p_1 + p'_1 + p_3$                 | $x_{00} + x_{01} + x_{10} + x_{11}$ |
| $\mathcal{W}_3$    | $p_1 + p_4 + p'_1 + p'_4 + p_{14}$ | $x_{00} + x_{01} + x_{10} + x_{12}$ |
| $\mathcal{W}_4$    | $p_1 + p_4 + p'_1 + p'_4$          |                                     |
| $\mathcal{W}_5$    | $p_1 + p_{14} + p_4$               | $x_{00} + x_{01} + x_{02}$          |
| $\mathcal{W}_6$    | $p_1 + p_3$                        |                                     |
| $\mathcal{W}_7$    | $p_1 + p_2 + p_3 + p_5$            |                                     |
| $\mathcal{W}_8$    | $p_1 + p_4$                        | $x_{00} + x_{01}$                   |
| $\mathcal{W}_9$    | $p_1 + p_{14} + p'_4$              |                                     |
| $\mathcal{W}_{10}$ | $p'_1 + p_3 + p_5 + p_{14}$        | $x_{00} + x_{01} + x_{10}$          |
| $\mathcal{W}_{11}$ | $p_1 + p_3 + p_5 + p'_1$           | $x_{00} + x_{01} + x_{10} + x_{02}$ |
| $\mathcal{W}_{12}$ | $p_1 + p_3 + p_5$                  |                                     |
| $\mathcal{W}_{13}$ | $p_1 + p'_1 + p_{14} + p_4$        |                                     |
| $\mathcal{W}_{14}$ | $p_1 + p'_1 + p_4$                 |                                     |

3-local structure, whose properties as a Lie algebra will be studied in the next section. The notation in  $B$  is described there. For the moment, we note that all  $x_{ij}$  belong to the set  $\mathcal{N}$  that we shall define shortly.

**Remark** The information in Table 3 and the earlier observations about geometric hyperplanes of weight at most 2 provide upper and lower bounds for the weights of the elements of  $V$ . To show that the weights are as stated in Table 2, it suffices to show that the elements of  $\mathcal{W}_7$  do in fact have weight only 3 and that nothing of weight 3 belongs to  $\mathcal{W}_3 \cup \mathcal{W}_4 \cup \mathcal{W}_{11} \cup \mathcal{W}_{13}$ . This can be done in a straightforward, but tedious, way by computing  $S(p) + S(q) + S(r)$  for representatives  $(p, q, r)$  from each  $G$ -orbit on  $\mathcal{P} \times \mathcal{P} \times \mathcal{P}$ .

Let  $T \in \text{Syl}_3(G)$  and write  $T_0 = Z(T)$ . The set  $\mathcal{N}$  of fixed points of  $T_0$  has size 9. There is a faithful action of  $N(T)/T_0$  on  $\mathcal{N}$ . The normal subgroup  $T/T_0$  acts regularly because  $T_0$  is a Sylow subgroup of every point stabilizer that contains it. We can regard  $T/T_0$  as a vector space over  $\mathbb{F}_3$ , or (by imposing the usual line structure) as an affine plane. Moreover,  $\mathcal{N}$  inherits this structure. We have  $N(T) = T : Q$ , where  $Q$  acts faithfully on  $T/T_0$  as a group of linear transformations. In fact  $|Q| = 16$ , so  $Q \cong QD_{16}$ , a Sylow 2-subgroup of  $GL_2(3)$ , which acts transitively on the set of 1-spaces of  $T/T_0$ . Thus  $T/T_0$  acts 2-transitively on the points of the plane. It follows that, in the collinearity graph of  $\mathcal{H}^*$ , the distance between distinct points of  $\mathcal{N}$  does not vary. It can be seen that this distance is 3 by noting that any group element fixing a pair of points at distance 2 also fixes the point adjacent to the original pair. Furthermore, if  $p, q \in \mathcal{N}$  and  $r = p \# q$  then

$r \in \mathcal{N}$  because  $T_0 \subseteq G_{p,q} \subseteq G_r$ . This implies that the lines of  $\mathcal{N}$ , as an affine plane, are the subsets of  $\mathcal{N}$  that are regulus triples.

Regard  $\mathcal{N}$  as a subset of  $V$  (consisting of element of  $\mathcal{W}_1$ ), and let  $B$  be the subspace spanned by  $\mathcal{N}$ . The  $N(T)$ -orbits on  $B$  can be described in terms of the geometry of  $\mathcal{N}$ , and this provides a relatively convenient way to describe a substantial portion of the  $G$ -orbits on  $V$ . For any  $\mathcal{M} \subseteq \mathcal{N}$ , let  $S(\mathcal{M}) = \sum_{p \in \mathcal{M}} S(p)$ . These geometric hyperplanes are the members of  $B$ .

We show first that  $S(\mathcal{N})$  is 0, the zero element  $\mathcal{P}$  of  $V$ . Observe that, from the definition of sums in  $V$ , a point  $r$  lies in  $S(\mathcal{N})$  if it is at maximal distance from an even number of points in  $\mathcal{N}$ . Since the points of  $\mathcal{N}$  lie at mutual distance 3,  $\mathcal{N} \subseteq S(\mathcal{N})$ . Also, if  $p$  and  $p'$  are distinct elements of  $\mathcal{N}$  then  $\Gamma_{\leq 1}(p) \cap \Gamma_{\leq 1}(p') = \emptyset$ . Since  $|\mathcal{P}| = 63 = |\mathcal{N}| \cdot |\Gamma_{\leq 1}(p)|$ , it follows that  $\{\Gamma_{\leq 1}(p) : p \in \mathcal{N}\}$  is a partition of  $\mathcal{P}$ . If  $q \in \mathcal{P} \setminus \mathcal{N}$  then  $\Gamma_1(q)$  must contain exactly one point in  $\mathcal{N}$ , and each of the four points on the two lines through  $q$  disjoint from  $\mathcal{N}$  must be collinear with a unique point of  $\mathcal{N}$ . Therefore  $q \notin S(p)$  for exactly four  $p \in \mathcal{N}$ , which implies that  $q \in S(\mathcal{N})$ . It follows that  $\mathcal{P} \subseteq S(\mathcal{N})$  — i.e.,  $S(\mathcal{N}) = 0$ .

Thus every element of  $B$  can be written as  $S(\mathcal{M})$  where  $\mathcal{M} \subseteq \mathcal{N}$  and  $|\mathcal{M}| \leq 4$ . We shall show that this expression is unique. If  $S(\mathcal{M}) = S(\mathcal{M}')$ , where  $\mathcal{M}$  and  $\mathcal{M}'$  are distinct subsets of  $\mathcal{N}$  of size at most 4, we may assume that these sets are disjoint. The complement in  $\mathcal{N}$  of their union is a set  $\mathcal{M}''$  with  $S(\mathcal{M}'') = 0$ . It is not hard to eliminate that possibility by examining the geometric hyperplanes of the form  $S(\mathcal{M})$ ,  $\mathcal{M} \subseteq \mathcal{N}$  (see Table 4). It follows from this analysis that  $B$  has dimension 8.

The orbits of  $N(T)$  on  $B$  are now easily determined by examining the corresponding orbits on subsets of  $\mathcal{N}$ . Since  $N(T)$  is 2-transitive on  $\mathcal{N}$ , it has a single orbit  $\mathcal{O}_1$  of length 9 on  $\mathcal{N}$  and a single orbit  $\mathcal{O}_2$  of length 36 on the 2-element subsets of  $\mathcal{N}$ . On the 3-element subsets there are two orbits  $\mathcal{O}_{3A}$ ,  $\mathcal{O}_{3B}$ , of respective lengths 12 and 72, corresponding to the lines of the affine plane and to the other 3-element subsets. Every line of the affine plane is contained in 6 distinct 4-sets, and the 72 such 4-sets form an  $N(T)$ -orbit  $\mathcal{O}_{4C}$ . The remaining 4-sets lie in two distinct  $N(T)$ -orbits. The complement of such a 4-set is the union of two non-parallel lines. Since  $Q$  is a 2-group the action of  $Q$  on the four lines through its fixed point is imprimitive, and we can define two intersecting lines in  $\mathcal{N}$  to be *perpendicular* if they lie in the same set of imprimitivity for the stabilizer of their common point. The 4-sets whose complementary lines are perpendicular form an orbit  $\mathcal{O}_{4A}$  of length 18, and the 4-sets whose complementary lines are not perpendicular form an orbit  $\mathcal{O}_{4B}$  of length 36.

This shows that  $N(T)$  has seven orbits on  $B$ . Geometric hyperplanes in distinct orbits have different parameters, so they are not fused in  $G$ . In Table 3 we list these  $N(T)$ -orbits together with the corresponding  $G$ -orbits,

TABLE 4:  $N(T)$ -orbits on  $B$ 

| $N(T)$ -orbit      | $N(T)$ -stabilizer | size | $G$ -orbit         | $G$ -stabilizer         |
|--------------------|--------------------|------|--------------------|-------------------------|
| $\mathcal{O}_1$    | $3 \times QD_{16}$ | 9    | $\mathcal{W}_1$    | $2^{1+4} : S_3$         |
| $\mathcal{O}_2$    | $D_{12}$           | 36   | $\mathcal{W}_8$    | $2^{1+2} : S_3$         |
| $\mathcal{O}_{3A}$ | $S_3 \times S_3$   | 12   | $\mathcal{W}_5$    | $S_3 \times S_3$        |
| $\mathcal{O}_{3B}$ | $S_3$              | 72   | $\mathcal{W}_{10}$ | $S_3$                   |
| $\mathcal{O}_{4A}$ | $D_{24}$           | 18   | $\mathcal{W}_2$    | $4 \circ 2^{1+2} : S_3$ |
| $\mathcal{O}_{4B}$ | $D_{12}$           | 36   | $\mathcal{W}_3$    | $D_{12}$                |
| $\mathcal{O}_{4C}$ | $S_3$              | 72   | $\mathcal{W}_{11}$ | $S_3$                   |

and the  $N(T)$ - and  $G$ -stabilizers. With the exception of the orbit  $\mathcal{O}_{3A}$  all of the  $G$ -stabilizers are contained in some conjugate of  $G_p$ .

For completeness, we note the following properties of structures associated with the Sylow 7-subgroup of  $G$ , even though they will play no further role. Let  $S \in \text{Syl}_7(G)$ . Then  $S$  has 9 regular orbits on  $\mathcal{P}$ , and  $N(S)$  acts transitively on the three non-trivial  $S$ -invariant geometric hyperplanes. Each of them belongs to  $\mathcal{W}_7$  and has stabilizer of order 14.  $N(S)$  has three orbits on  $\mathcal{P}$ , each of length 21. One of these is invariant under the unique  $PGL_2(7)$  subgroup containing  $S$ . The points of this orbit together with the lines they contain form a subhexagon of order  $(2, 1)$ . The other two  $N(S)$ -orbits are stabilized only by  $N(S)$ .

## 6. Relations between $\mathcal{H}^*$ and the Lie algebra $\mathfrak{g}_2(2)$

The group  $G = G_2(2)$  acts on  $\mathfrak{g}_2(2)$  as a group of automorphisms defined with respect to some fixed Chevalley basis. The necessary background can be found in Carter [1]. The Lie algebra  $\mathfrak{g}_2(2)$  is unusual in that it has a considerably larger automorphism group (isomorphic to  $Sp_6(2)$ ). Some of the consequences will be examined below. We shall also see, from a geometric point of view, both classical and ‘sporadic’ subalgebras of  $\mathfrak{g}_2(2)$ . One useful observation is that the subspace of  $\mathfrak{g}_2(2)$  fixed by a subgroup of  $G$  is a Lie subalgebra.

Recall that  $U = U(\mathcal{H}^*)$  and  $V = V(\mathcal{H}^*)$  are adjoint modules for  $G$ , so they can be endowed with a  $G$ -invariant Lie bracket. By [7], each of these modules supports a unique nontrivial  $G$ -invariant algebra operation. Moreover, explicit formulas for these operations are given in parts (b) and (c) of Lemma 2.2. One need only check that the operation on  $V$  is nontrivial, and that the one on  $U$  arises from it via the isomorphism  $U \cong V$  defined in Lemma 2.1(b). Thus the Lie brackets on  $U$  and  $V$  have natural geometric definitions.

We shall often work with the algebra defined on  $U$ . For convenience, we record the following formula for certain commutators in  $U$ . It can be verified by a straightforward computation which is most easily carried out in the module  $V$  of geometric hyperplanes of  $\mathcal{H}^*$ .

**Recipe 6.1** *Let  $p$  and  $q$  be points of the embedded image of  $\mathcal{H}$  in  $U$ .*

- (a) *If  $p$  and  $q$  are collinear then  $[p, q] = 0$ .*
- (b) *If  $d(p, q) = 2$  then  $[p, q] = r$ , where  $r$  is the unique point collinear with  $p$  and  $q$ .*
- (c) *If  $d(p, q) = 3$  then  $[p, q] = p + q + p\#q$ .*

### The standard Chevalley basis

A Chevalley basis  $\mathcal{B}$  of  $U$  is a set of 14 elements satisfying the usual commutator relations, as described for example in [1]. However, for our purposes, we shall often regard  $\mathcal{B}$  as an ordered set. Then  $\text{Aut}(\mathfrak{g}_2(2))$  acts regularly on the set of all such bases. Given any  $\mathcal{B}$ , Chevalley's construction defines a group of algebra automorphisms of  $U$  that is isomorphic to  $G_2(2)$ . If this group coincides with the one induced by  $G$  on  $U$ , we say that  $\mathcal{B}$  is a standard Chevalley basis. It is easy to see that  $G$  acts regularly on the set of all standard Chevalley bases.

The elements of  $V$  that belong to some standard Chevalley basis can be identified geometrically. It is clear from the detailed description in [7] of the adjoint embedding of  $\mathcal{H}^*$  that the  $G$ -orbit consisting of the long root vectors is  $\mathcal{W}_1$  in the module  $V$  of geometric hyperplanes of  $\mathcal{H}^*$ , or the image of  $\mathcal{P}$  in the universal embedding space  $U$ . One could also use the weight theory of algebraic groups to reach the same conclusion, since vectors in  $\mathcal{W}_1$  and long root elements are stabilized by parabolic subgroups. Thus the long root vectors of  $U$  (or of  $V$ ) are in natural one-to-one correspondence with the elements of  $\mathcal{P}$ .

Lemma 5.1 implies that the standard Chevalley bases for  $U$  are in one-to-one correspondence with the ordered 6-circuits. To make this correspondence explicit, we can construct a Chevalley basis for  $U$  from our 6-circuit  $p_1, \dots, p_6$  by first constructing a Chevalley basis for the  $\mathfrak{sl}_3(2)$  subalgebra generated by  $p_1, \dots, p_6$ . Letting  $h_j = [p_j, p_{j+3}]$ ,  $j = 1, 2$ , it is easy to verify that  $\{h_1, h_2\} \cup \{s_i : i = 1, \dots, 6\}$  is a Chevalley basis for a long root  $\mathfrak{sl}_3(2)$  subalgebra. This basis for the  $\mathfrak{sl}_3(2)$  subalgebra completely determines the remaining six elements needed to complete it to a Chevalley basis in which the long root elements are the elements of the given 6-circuit. Each of the six short root elements of the basis is the unique non-trivial element of  $U$  that centralizes  $p_i$  for some four consecutive values of  $i$ , read modulo 6. That there is a unique such element of  $U$  follows directly from the Chevalley commutator

relations. We shall soon show how these elements can be explicitly located using the geometry of  $\mathcal{H}^*$ .

The previous paragraph shows that sums of the form  $p + q + p \# q$  are semisimple elements of  $U$ . (This can also be done directly, using the recipe.) Under the canonical identification of  $U$  with  $V$ , these elements lie in the  $G$ -orbit  $\mathcal{W}_5$ . For each point  $p \in \mathcal{P}$ , the 32 points in  $\Gamma_3(p)$  form 16 pairs of the form  $r, p \# r$ . Since each regulus triple contains three points, this implies that the number of regulus triples is  $63 \cdot 16 / 3 = 336 = |\mathcal{W}_5|$ . This sets up a bijection between elements of  $\mathcal{W}_5$  and regulus triples.

Using the notation from Section 5, we can specify the element that centralizes  $p_1, p_2, p_3, p_4$ . In particular, we claim that  $x = p_1 + p_2 + p_3 + p_4 + p'_1 + p'_4 + p_{14} + p'_{14}$  centralizes  $\langle p_1, p_2, p_3, p_4 \rangle$ . The recipe shows that  $p_2$  centralizes  $p_1, p_2, p'_1$  and  $p_3$ , as well as  $p_4 + p'_4$  and  $p_{14} + p'_{14}$ . Consequently  $[p_2, x] = 0$ . Similarly,  $[p_3, x] = 0$ . On the other hand,  $p_1$  centralizes  $p_1 + p_4 + p_1 \# p_4$  and  $p_1 + p'_4 + p_1 \# p'_4$ ,  $p_1 + p'_1 + p_3$ , and  $p_2$ . Since  $p_1 \# p_4 = p_{14}$  and  $p_1 \# p'_4 = p'_{14}$ , it follows that  $[p_1, x] = 0$ . The argument that  $[p_4, x] = 0$  is similar, and, in fact, it is not difficult to see that  $[p_{14}, x] = 0$  as well. It is a straightforward matter to compute that  $x \in \mathcal{W}_4$ .

The previous paragraph implies that  $x$  centralizes all 9 points on the lines determined by  $p_1, p_2, p_3, p_4, p_2 * p_3 = p_2 + p_3$  and  $p_{14}$ . These are precisely the fixed points of  $t$  on  $V$ , and it follows that  $C_G(t) \subseteq G_x$ . It can be shown, by examining the structure of a geometric hyperplane in  $\mathcal{W}_4$ , that  $G_x$  does not properly contain  $C_G(t)$ . Thus  $t$  fixes a unique member of  $\mathcal{W}_4$ , as claimed in Section 5. This sets up a bijection between  $\mathcal{W}_4$  and the  $G$ -conjugates of  $t$ .

#### An 8-dimensional algebra associated with the 3-local structure of $G$

In the notation of the previous section,  $\mathcal{N}$  is a set fixed by the center of a Sylow 3-subgroup of  $G$ . It consists of nine points at mutual distance 3. It was shown above that if  $p, q, r$  is a regulus triple, then  $p + q + r$  is a semisimple element of  $U$ . Next note that  $[p, q] = p + q + r$  whenever  $\{p, q, r\}$  is a line of  $\mathcal{N}$ . It follows that (in  $U$  or in  $V$ ) the subspace  $B$  spanned by  $\mathcal{N}$  is closed under the Lie bracket. For clarity we let  $\mathfrak{b}$  denote the corresponding Lie algebra.

This leads us to consider the following direct method of constructing a Lie algebra from a 9-point affine plane  $\mathcal{A}$ . Let  $\tilde{\mathfrak{a}}$  be an  $\mathbb{F}_2$ -vector space with basis  $x_a$ ,  $a \in \mathcal{A}$ . Set  $[x_a, x_a] = 0$  and, for distinct  $a, b$  in  $\mathcal{A}$ , set  $[x_a, x_b] = x_a + x_b + x_c$ , where  $\{a, b, c\}$  is a line in  $\mathcal{A}$ . Then this operation extends uniquely to a bilinear multiplication on  $\tilde{\mathfrak{a}}$ . Define  $z = \sum \{x_a : a \in \mathcal{A}\}$ , and let  $\mathfrak{a} = \tilde{\mathfrak{a}} / \langle z \rangle$ . Since  $[z, x] = 0$  for all  $x \in \tilde{\mathfrak{a}}$ , the multiplication defined on  $\tilde{\mathfrak{a}}$  determines one on  $\mathfrak{a}$ . We list some of its properties.

#### Proposition 6.2

- (a)  $\mathfrak{a}$  is a simple Lie algebra of dimension 8.

- (b) The group  $\text{Aff}_2(3) \cong 3^2:GL_2(3)$  acts faithfully on  $\mathfrak{a}$ .
- (c)  $\mathfrak{a} \not\cong \mathfrak{sl}_3(2)$ .
- (d)  $\mathfrak{a} \otimes \mathbb{F}_4 \cong \mathfrak{sl}_3(4)$  (so  $\mathfrak{a}$  is a 'form' of  $\mathfrak{sl}_3(2)$ ).
- (e)  $\mathfrak{g}_2(2)$  has subalgebras isomorphic to  $\mathfrak{a}$ .

**Proof** Evidently  $\mathfrak{a}$  has dimension 8, and  $[x, x] = 0$  for all  $x \in \mathfrak{a}$ . To check the Jacobi identity, it suffices to consider the images in  $\mathfrak{a}$  of distinct elements  $x_a, x_b, x_c$  from the given basis of  $\tilde{\mathfrak{a}}$ . In order to avoid the trivial case in which  $[[x_a, x_b], x_c]$  and other similar terms vanish, we shall suppose that  $\{a, b, c\}$  is not a line of  $\mathcal{A}$ . Let  $\{c, d, e\}$  be the line through  $c$  parallel to the line through  $a$  and  $b$ . A short computation shows that, in  $\tilde{\mathfrak{a}}$ , we have  $[[x_a, x_b], x_c] = z + x_d + x_e$ , since we are working over  $\mathbb{F}_2$ . By permuting  $a, b, c$  cyclically, and adding the three formulas thus obtained, we obtain  $z$  as the sum. This verifies that  $\mathfrak{a}$ , but not  $\tilde{\mathfrak{a}}$ , satisfies the Jacobi identity. Thus  $\mathfrak{a}$  is a Lie algebra. Its simplicity will follow from (d), since  $\mathfrak{sl}_3(4)$  is simple.

From the symmetry of the construction it is clear that any nontrivial automorphism of the affine plane  $\mathcal{A}$  induces a nontrivial action on the Lie algebra  $\mathfrak{a}$ . Thus (b) holds. This implies (c), because  $\mathfrak{sl}_3(2)$  does not admit the faithful action of a non-cyclic 3-group — see [5], for example.

To establish (d), regard  $\mathcal{A}$  as  $\{(\mu, \nu) : \mu, \nu \in \mathbb{F}_3\}$ . For  $\mu, \nu \in \mathbb{F}_3$ , set  $h_\mu = x_{(\mu, 0)} + x_{(\mu, 1)} + x_{(\mu, 2)}$ , and  $y_{\mu, \nu} = h_\mu + x_{(\mu, \nu)}$ . Then  $[h_\mu, h_\nu] = 0$ ,  $[h_\mu, y_{\mu, \nu}] = 0$ , and  $[h_\mu, y_{\mu', \nu}] = y_{\mu', \nu}$ ,  $\mu' \neq \mu$ . Also, when  $\mu, \mu', \mu''$  are distinct and  $\nu, \nu', \nu''$  are distinct,  $[y_{\mu, \nu}, y_{\mu, \nu'}] = h_\mu$ ,  $[y_{\mu, \nu}, y_{\mu', \nu}] = y_{\mu'', \nu}$ ,  $[y_{\mu, \nu}, y_{\mu', \nu'}] = y_{\mu'', \nu''}$ .

Extending the scalars to  $\mathbb{F}_4 = \mathbb{F}_2(\omega)$ , we observe that  $\mathfrak{a} \otimes \mathbb{F}_4$  is isomorphic to  $\mathfrak{sl}_3(4)$ , because it has the following Chevalley basis:

$$\begin{aligned} h_\alpha &= h_1 \\ h_\beta &= h_2 \\ e_\alpha &= \omega y_{1,1} + \omega^2 y_{1,2} \\ e_{-\alpha} &= \omega^2 y_{1,1} + \omega y_{1,2} \\ e_\beta &= \omega y_{2,1} + \omega^2 y_{2,2} \\ e_{-\beta} &= \omega^2 y_{2,1} + \omega y_{2,2} \\ e_{\alpha+\beta} &= \omega^2 y_{0,1} + \omega y_{0,2} \\ e_{-\alpha-\beta} &= \omega y_{0,1} + \omega^2 y_{0,2}. \end{aligned}$$

For (e) we work in  $U$ , with the usual Lie bracket giving it the structure of  $\mathfrak{g}_2(2)$ . Recall that  $B$ , the span of the 9-point set  $\mathcal{N}$ , supports a Lie subalgebra  $\mathfrak{b}$  of  $U$ . It is clear from the way in which pairs of elements of  $\mathcal{N}$  multiply that

there is an algebra epimorphism  $f : \tilde{\mathfrak{a}} \rightarrow \mathfrak{b}$ . Because  $\mathfrak{b}$  satisfies the Jacobi identity, the computation used in (a) implies that  $f(z) = 0$ . This also follows from the fact, proved in the previous section, that  $\sum_{p \in \mathcal{N}} S(p) = 0$ . We now obtain a Lie algebra epimorphism  $\mathfrak{a} \rightarrow \mathfrak{b}$ , which is an isomorphism because  $\mathfrak{a}$  is simple, or because  $\mathfrak{b}$ , as well as  $\mathfrak{a}$ , is 8-dimensional.  $\square$

### Fusion in $\text{Aut}(\mathfrak{g}_2(2))$

It is an easy consequence of the Chevalley commutator relations for  $\mathfrak{g}_2(2)$  that the algebra  $V$  admits an automorphism that permutes the standard Chevalley basis differently from the permutations induced from the Weyl group. Let  $\Phi = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta), \pm(2\alpha + \beta), \pm(3\alpha + \beta), \pm(3\alpha + 2\beta)\}$  be a root system of type  $G_2$ , and let  $\{h_\alpha, h_\beta\} \cup \{e_\gamma, \gamma \in \Phi\}$  be a Chevalley basis for  $\mathfrak{g}_2(2)$ . For convenience, we write  $e_{ij}$  for  $e_{i\alpha + j\beta}$  and  $f_{ij}$  for  $e_{-i\alpha - j\beta}$ ,  $i, j \geq 0$ . Because the characteristic is 2, the Chevalley commutator relations imply that  $[e_{10}, e_{01}] = e_{11}$  and  $[e_{10}, e_{21}] = e_{31}$  while  $[e_{10}, e_{11}] = [e_{10}, e_{31}] = 0$ .

The involution  $\sigma : \mathfrak{g}_2(2) \mapsto \mathfrak{g}_2(2)$ , induced from the permutation

$$(e_{01}, e_{21})(f_{01}, f_{21})(e_{11}, e_{31})(f_{11}, f_{31})$$

of the unordered Chevalley basis, is therefore a Lie algebra automorphism. Since  $\sigma$  interchanges the long root vector  $e_{01} \in \mathcal{W}_1$  with the short root vector  $e_{21} \in \mathcal{W}_4$ , it is clear that  $\sigma \notin G$ , and even that  $\sigma$  does not normalize  $G$  because  $\mathcal{W}_1$  and  $\mathcal{W}_4$  have different sizes. Certain  $G$ -orbits fused by  $\sigma$ , with representatives that demonstrate the fusion, are listed in the following table.

TABLE 5: Fusion induced by  $\sigma$

| $G$ -orbit of $v$  | $v$                                 | $v^\sigma$                          | $G$ -orbit of $v^\sigma$ |
|--------------------|-------------------------------------|-------------------------------------|--------------------------|
| $\mathcal{W}_1$    | $e_{01}$                            | $e_{21}$                            | $\mathcal{W}_4$          |
| $\mathcal{W}_{14}$ | $e_{10} + e_{01}$                   | $e_{10} + e_{21}$                   | $\mathcal{W}_2$          |
| $\mathcal{W}_2$    | $e_{11} + e_{21}$                   | $e_{01} + e_{31}$                   | $\mathcal{W}_6$          |
| $\mathcal{W}_6$    | $e_{11} + e_{21} + e_{31}$          | $e_{11} + e_{01} + e_{31}$          | $\mathcal{W}_3$          |
| $\mathcal{W}_8$    | $h_2 + e_{01}$                      | $h_2 + e_{21}$                      | $\mathcal{W}_9$          |
| $\mathcal{W}_9$    | $h_2 + e_{10} + e_{21}$             | $h_2 + e_{10} + e_{01}$             | $\mathcal{W}_{13}$       |
| $\mathcal{W}_{12}$ | $e_{01} + e_{31} + f_{32}$          | $e_{21} + e_{11} + f_{32}$          | $\mathcal{W}_{11}$       |
| $\mathcal{W}_7$    | $e_{01} + e_{31} + e_{32} + f_{32}$ | $e_{21} + e_{11} + e_{32} + f_{32}$ | $\mathcal{W}_{10}$       |

Let  $G^*$  denote the full automorphism group of the Lie algebra  $(V, [\cdot, \cdot])$ . By considering the action of  $\text{ad } x$  for  $x \in V$ , it is not hard to see that the full list of  $G$ -orbits that fuse can be deduced from the information in

TABLE 6: Fusion of orbits in  $\text{Aut}(V, [\cdot, \cdot])$ 

| $G$ -orbits                                                                 | size of $G^*$ -orbit | $\dim[V, x]$ | action of $\text{ad } x$ |
|-----------------------------------------------------------------------------|----------------------|--------------|--------------------------|
| $\mathcal{W}_1 \cup \mathcal{W}_4$                                          | 315                  | 6            | nilpotent                |
| $\mathcal{W}_5$                                                             | 336                  | 8            | semisimple               |
| $\mathcal{W}_7 \cup \mathcal{W}_{10}$                                       | 2880                 | 12           | order 7 on $[V, x]$      |
| $\mathcal{W}_2 \cup \mathcal{W}_3 \cup \mathcal{W}_6 \cup \mathcal{W}_{14}$ | 3780                 | 10           | nilpotent                |
| $\mathcal{W}_{11} \cup \mathcal{W}_{12}$                                    | 4032                 | 12           | order 3 on $[V, x]$      |
| $\mathcal{W}_8 \cup \mathcal{W}_9 \cup \mathcal{W}_{13}$                    | 5040                 | 10           | order 2 on $[[V, x], x]$ |

Table 5. This list, together with properties of  $\text{ad } x$  that allow the  $G^*$ -orbits to be distinguished, is given in Table 6. In fact  $G^* \cong Sp_6(2)$  — see [5], for example. Therefore  $G$  is maximal in  $G^*$ , and  $G^* = \langle G, \sigma \rangle$ .

### Nonstandard Chevalley bases for $V$

Since  $G^*$  acts regularly on the collection of all (ordered) Chevalley bases of  $V$ , many of them are non-standard. It was stated incorrectly in [6] that there are only three types of Chevalley bases, but actually the number of geometrically distinct bases is sufficiently large that a complete list would be of little interest.

There are, however, just two  $G$ -orbits of toral subalgebras (2-dimensional algebras generated by the semisimple elements of some Chevalley basis). We indicate briefly the argument that establishes this. Table 6 shows that every semisimple element of  $V$  belongs to  $\mathcal{W}_5$ . There is a one-to-one correspondence between elements of this  $G$ -orbit and regulus triples in  $\mathcal{H}^*$ . A computer calculation within the Lie algebra centralizer of an element of  $\mathcal{W}_5$  shows that that element belongs to a unique nonstandard toral subalgebra and to nine distinct standard toral subalgebras. In particular,  $G$  has exactly two orbits (fused by  $G^*$ ) on the set of toral subalgebras. Thus, from the regulus triples, each toral subalgebra determines a 9-element subset of  $\mathcal{P}$ . For a standard toral subalgebra, this set contains a 6-circuit, while the set determined by a toral subalgebra of the 8-dimensional algebra  $\mathfrak{b}$  consists of points at mutual distance 3.

### REFERENCES

- [1] R. Carter, *Simple Groups of Lie Type*, Wiley, New York-London-Sydney, 1972.
- [2] A. Cohen and J. Tits, *On generalized hexagons and a near octagon whose lines have three points*, Europ. J. Combinatorics **6** (1985), 13–27.

- [3] J. H. Conway, R. T. Curtis, S. P. Norton, R. P. Parker, R. A. Wilson, *Atlas of Finite Groups*, Clarendon Press, Oxford, 1985.
- [4] P. Dembowski, *Finite Geometries*, Springer, New York, 1968.
- [5] D. Frohardt and R. Griess, *Automorphisms of modular Lie algebras*, Nova J. of Algebra and Geometry, to appear.
- [6] D. Frohardt and P. Johnson, *Geometric hyperplanes in generalized  $n$ -gons with 3 points on a line*, pp. 175–180 in: 'Coding Theory, Design Theory and Graph Theory,' Proceedings of Marshall Hall Memorial Conference (eds.: K. Arasu, D. Jungnickel), Wiley, 1992.
- [7] D. Frohardt and P. Johnson, *On the uniqueness of the  $G_2$  algebra in characteristic 2*, submitted.
- [8] D. Frohardt and S. Smith, *Universal embeddings for the  ${}^3D_4(2)$  hexagon and  $J_2$  near-octagon*, Europ. J. Combinatorics, to appear.
- [9] M. A. Ronan, *Embeddings and Hyperplanes of Discrete Geometries*, Europ. J. Combinatorics 8 (1987), 179–185.
- [10] H. Völklein, *On the geometry of the adjoint representation of a Chevalley group*, J. Algebra 127 (1989), 139–154.

Received: January 1993